

Manifestoberta Performance Report

Context Version 2024a (56Topics)

2024-07-31

Summary

- Performance was measured on 203 manifestos, which represent 200920 annotated quasi-sentences.
- Overall the model manages to assign the correct category to quasi-sentences in the test data set with an accuracy of 64.23%. In 81.27% of the cases, the true category is among the two most confident predictions of the model, and in 88.37% among the top three (Table 2).
- Lower macro averaged F1, Precision, and Recall reflect problems with some individual categories, especially rare/exotic categories like 102, 409, or 702 (Table 2 and Table 3).
- The overall distribution and frequency of individual category predictions is closely aligned with the true distribution of categories in our test data set. The model isn't systematically over- or under predicting specific codes (Table 3).
- The model performs considerably well in all countries/languages present in the test data set, with the lowest accuracy value of 51.08% in Italy (Table 4).
- Probability estimates of the model are well calibrated and properly reflect the likelihood of a right prediction. If the model reports a confidence of 95% or higher (which happened on 15.56% of all quasi-sentences in our test set) it was, in fact, right in 94.67% of those cases (Table 5).
- True rile values and rile values calculated based on model predictions are strongly correlated (Plot 1). However, there are a few cases where large differences occurred (over 30) between true rile and model rile (Plot 2).

Usage Recommendations

Given the capabilities of the model, a number of use cases are conceivable:

- The model's probability estimates make it possible to automatically predict a subset of a document and focus manual labeling efforts only on the more uncertain cases. For instance, if a cutoff of 80% confidence is chosen, almost 40% of the sentences to be coded could be automatically classified with an accuracy of at least 73%.
- To support manual coding, the model predictions can be used as code suggestions. This narrows down the choices during coding and speeds up the process significantly without sacrificing quality, considering an accuracy of 88% for the top 3 suggestions of the model and 94% for the top 5 suggestions.
- It is also conceivable to code documents completely automatically using the model. However, it might then be advisable to at least manually validate parts of the codings for subsequent analysis and/or to robustly integrate the automatically generated codes into the analysis (e.g. multiple variations of random substitutions between top picks, bootstrapping).

Model History

The model is regularly updated. All versions are available on Hugging Face (<https://huggingface.co/manifesto-project>). The performance of the 2024-1-1 model is comparable to the 2023-1-1 model, with only minor difference (see Table 1). It should be noted that the test datasets for these models differ, making direct comparisons less straightforward. As the performance variations are minimal, it is difficult to draw firm conclusions about whether the newer model performs better or worse.

Model	Accuracy	Top2_Acc	Top3_Acc	Precision	Recall	F1macro	MCC	Cross_Entropy
2024-1-1	0.64	0.81	0.88	0.55	0.52	0.53	0.63	1.15
2023-1-1	0.64	0.81	0.88	0.54	0.52	0.53	0.62	1.15

Table 1: Classification Results - Comparison between Models

Results 2024-1-1

Accuracy	0.64
Top2_Acc	0.81
Top3_Acc	0.88
Precision	0.55
Recall	0.52
F1macro	0.53
MCC	0.63
Cross-Entropy	1.15

Table 2: Classification Results - Overall

Category	Precision	Recall	F1	n(%)	n_predicted(%)
101	0.46	0.53	0.50	0.29%	0.33%
102	0.57	0.49	0.53	0.07%	0.06%
103	0.48	0.43	0.46	0.26%	0.23%
104	0.74	0.79	0.76	1.55%	1.66%
105	0.59	0.74	0.66	0.34%	0.42%
106	0.54	0.67	0.60	0.34%	0.43%
107	0.63	0.66	0.65	2.35%	2.47%
108	0.64	0.67	0.65	1.24%	1.31%
109	0.47	0.31	0.37	0.16%	0.10%
110	0.61	0.62	0.62	0.43%	0.44%
201	0.58	0.59	0.59	2.20%	2.23%
202	0.65	0.59	0.62	3.57%	3.24%
203	0.45	0.38	0.41	0.18%	0.15%
204	0.54	0.58	0.56	0.21%	0.22%
301	0.63	0.65	0.64	2.01%	2.07%
302	0.42	0.23	0.29	0.17%	0.09%
303	0.59	0.56	0.57	4.35%	4.12%
304	0.73	0.64	0.68	1.55%	1.37%
305	0.57	0.54	0.56	2.06%	1.98%
401	0.46	0.35	0.39	1.11%	0.84%
402	0.57	0.57	0.57	2.50%	2.51%
403	0.54	0.50	0.52	3.19%	2.97%
404	0.45	0.28	0.34	0.59%	0.36%
405	0.41	0.28	0.33	0.25%	0.18%
406	0.42	0.36	0.39	0.40%	0.34%
407	0.53	0.51	0.52	0.47%	0.46%
408	0.38	0.25	0.30	1.52%	1.00%
409	0.22	0.12	0.15	0.28%	0.14%
410	0.57	0.53	0.55	2.00%	1.86%
411	0.69	0.76	0.72	8.50%	9.27%
412	0.42	0.18	0.25	0.63%	0.26%
413	0.52	0.69	0.59	0.37%	0.48%
414	0.59	0.58	0.59	1.22%	1.20%
415	0.47	0.37	0.41	0.13%	0.10%
416	0.61	0.44	0.51	3.07%	2.20%
501	0.68	0.82	0.74	5.72%	6.87%
502	0.76	0.86	0.81	3.24%	3.63%
503	0.62	0.61	0.61	6.00%	5.94%
504	0.70	0.77	0.73	9.47%	10.40%
505	0.56	0.39	0.46	0.68%	0.48%
506	0.75	0.82	0.78	5.54%	6.08%
507	0.47	0.19	0.27	0.12%	0.05%
601	0.57	0.53	0.55	1.55%	1.43%
602	0.35	0.31	0.33	0.30%	0.26%
603	0.64	0.69	0.67	1.10%	1.19%
604	0.56	0.62	0.59	0.49%	0.55%
605	0.68	0.75	0.71	3.94%	4.32%
606	0.56	0.48	0.51	1.48%	1.27%
607	0.64	0.66	0.65	1.44%	1.50%
608	0.46	0.47	0.47	0.26%	0.26%
701	0.66	0.67	0.66	3.53%	3.53%
702	0.46	0.37	0.41	0.09%	0.07%
703	0.79	0.86	0.82	3.06%	3.32%
704	0.45	0.26	0.33	0.37%	0.22%
705	0.42	0.25	0.32	0.86%	0.51%
706	0.41	0.35	0.38	1.19%	1.03%

Table 3: Classification Results - Categories

Country	Accuracy	Precision	Recall	F1	N
Argentina	64.57%	0.54	0.57	0.55	1,160
Armenia	63.89%	0.57	0.60	0.65	108
Australia	71.89%	0.55	0.56	0.59	6,215
Austria	61.81%	0.46	0.51	0.49	7,798
Belgium	56.50%	0.46	0.43	0.43	16,221
Bolivia	53.79%	0.39	0.38	0.40	924
Bosnia-Herzegovina	53.09%	0.44	0.41	0.41	1,746
Bulgaria	58.61%	0.50	0.48	0.50	1,005
Canada	58.74%	0.48	0.43	0.45	3,551
Chile	54.47%	0.41	0.47	0.43	6,405
Colombia	70.89%	0.41	0.54	0.48	7,534
Costa Rica	72.75%	0.58	0.56	0.57	6,796
Croatia	77.55%	0.62	0.70	0.70	3,194
Cyprus	65.30%	0.48	0.48	0.50	2,977
Czech Republic	63.17%	0.54	0.50	0.51	5,009
Denmark	64.81%	0.54	0.45	0.47	2,376
Dominican Republic	71.66%	0.58	0.59	0.56	4,107
Ecuador	52.78%	0.48	0.42	0.49	360
Estonia	71.13%	0.58	0.52	0.57	2,865
Finland	67.29%	0.55	0.53	0.54	2,975
France	65.03%	0.48	0.50	0.51	2,236
Germany	61.05%	0.51	0.50	0.49	11,226
Greece	64.82%	0.52	0.52	0.52	1,424
Hungary	68.00%	0.58	0.52	0.55	3,650
Iceland	64.27%	0.61	0.55	0.59	375
Israel	71.50%	0.58	0.60	0.61	1,695
Italy	51.08%	0.40	0.34	0.35	7,606
Latvia	80.53%	0.73	0.69	0.85	113
Lithuania	71.34%	0.56	0.53	0.58	2,055
Mexico	53.41%	0.40	0.40	0.41	777
Montenegro	60.68%	0.57	0.52	0.56	562
Netherlands	61.92%	0.50	0.45	0.47	10,835
New Zealand	67.48%	0.55	0.50	0.52	8,819
North Macedonia	70.51%	0.61	0.50	0.56	9,116
Norway	65.43%	0.42	0.48	0.44	5,505
Panama	71.22%	0.55	0.57	0.57	952
Peru	72.28%	0.56	0.59	0.58	4,790
Poland	65.69%	0.54	0.56	0.58	822
Portugal	59.59%	0.49	0.43	0.43	7,290
Romania	63.32%	0.49	0.54	0.56	398
Russia	62.03%	0.61	0.57	0.57	1,164
Serbia	66.49%	0.65	0.58	0.60	382
Slovakia	64.69%	0.55	0.51	0.51	3,302
Slovenia	57.91%	0.43	0.44	0.47	1,668
South Africa	71.82%	0.55	0.56	0.62	802
South Korea	66.61%	0.52	0.50	0.62	626
Spain	70.09%	0.57	0.54	0.55	9,514
Sweden	66.26%	0.55	0.48	0.53	2,398
Switzerland	59.07%	0.55	0.56	0.59	215
Turkey	69.72%	0.56	0.56	0.57	6,601
Ukraine	58.31%	0.49	0.47	0.48	439
United Kingdom	64.21%	0.57	0.54	0.55	2,998
United States	57.67%	0.47	0.48	0.46	3,657
Uruguay	54.22%	0.38	0.39	0.37	3,582

Note: N relates to the number of quasi-sentences in the test dataset.

Table 4: Classification Results - Countries

Parfam	Accuracy	Precision	Recall	F1	N
10	60.80%	0.48	0.45	0.45	17,701
20	62.19%	0.47	0.46	0.45	25,211
30	65.55%	0.53	0.51	0.51	39,639
40	63.87%	0.53	0.51	0.52	28,942
50	65.32%	0.56	0.51	0.53	22,771
60	66.51%	0.56	0.52	0.52	31,692
70	64.30%	0.51	0.52	0.51	6,810
80	72.69%	0.52	0.57	0.53	5,390
90	62.40%	0.51	0.46	0.49	8,255
95	56.97%	0.50	0.45	0.45	11,726
98	74.80%	0.55	0.53	0.56	2,377
999	71.43%	0.65	0.63	0.66	406

Note: N relates to the number of quasi-sentences in the test dataset.

Table 5: Classification Results - Parfam

prob_estimates	accuracy	n(%)	cum_n(%)
> 95%	94.67%	15.56%	15.56%
90%-95%	85.62%	10.36%	25.93%
85%-90%	78.46%	8.13%	34.05%
80%-85%	73.70%	7.07%	41.12%
75%-80%	68.42%	6.34%	47.46%
70%-75%	64.03%	6.13%	53.58%
65%-70%	59.07%	5.98%	59.56%
60%-65%	55.89%	6.03%	65.59%
55%-60%	50.60%	6.22%	71.82%
50%-55%	46.68%	6.42%	78.23%
45%-50%	43.27%	6.25%	84.49%
40%-45%	38.82%	5.15%	89.63%
35%-40%	34.41%	4.21%	93.84%
30%-35%	29.92%	3.08%	96.92%
25%-30%	23.60%	1.91%	98.83%
20%-25%	19.38%	0.92%	99.75%
15%-20%	14.50%	0.23%	99.98%
10%-15%	15.15%	0.02%	100.00%
5%-10%	100.00%	0.00%	100.00%

Table 6: Model Calibration - Probability Groups

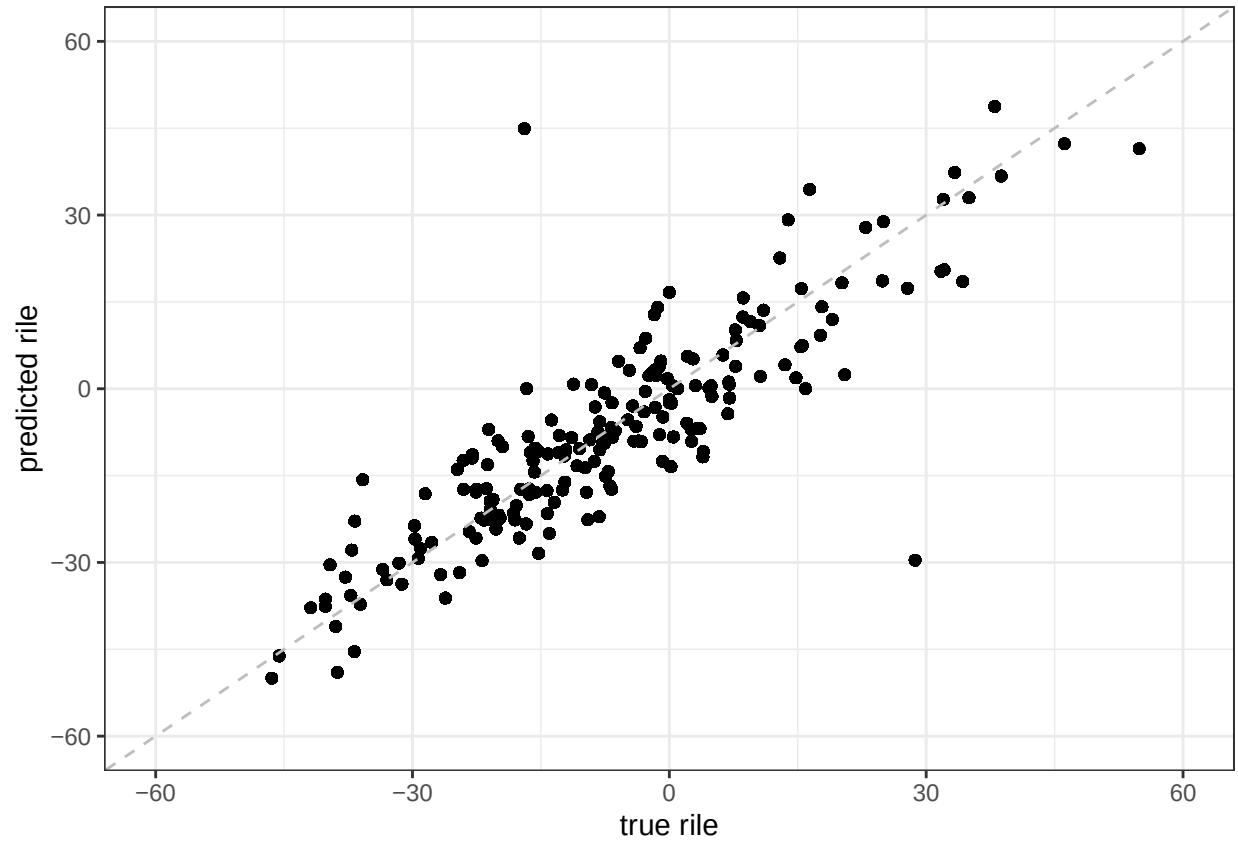


Figure 1: True rile values vs. predicted rile values

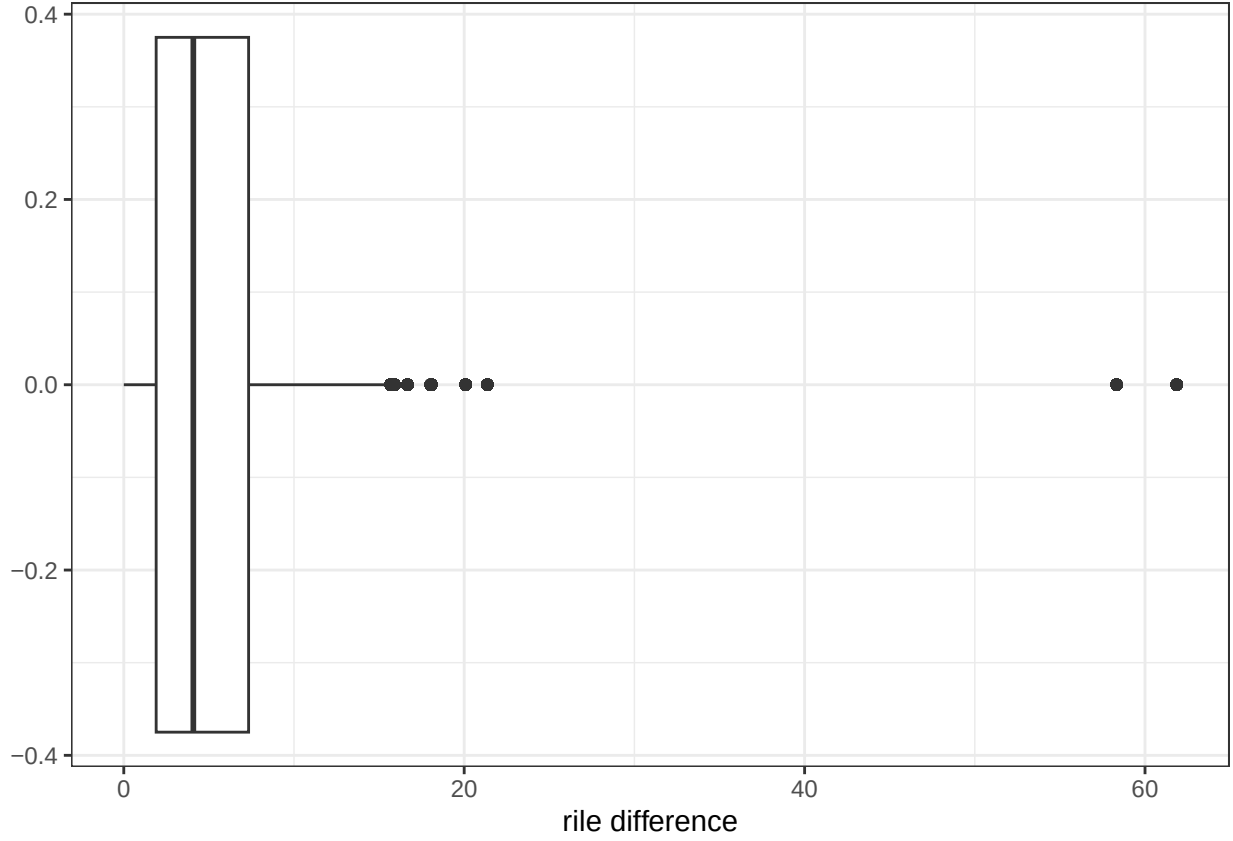


Figure 2: Absolute rile differences between true and predicted cmp codes

Results 2023-1-1

Accuracy	0.64
Top2_Acc	0.81
Top3_Acc	0.88
Precision	0.54
Recall	0.52
F1macro	0.53
MCC	0.62
Cross-Entropy	1.15

Table 7: Classification Results - Overall

Category	Precision	Recall	F1	n(%)	n_predicted(%)
101	0.50	0.48	0.49	0.30%	0.29%
102	0.56	0.61	0.58	0.09%	0.10%
103	0.51	0.36	0.42	0.28%	0.20%
104	0.78	0.81	0.79	1.57%	1.64%
105	0.69	0.70	0.69	0.34%	0.34%
106	0.59	0.57	0.58	0.33%	0.32%
107	0.68	0.66	0.67	2.24%	2.17%
108	0.66	0.68	0.67	1.20%	1.24%
109	0.52	0.39	0.45	0.17%	0.13%
110	0.63	0.68	0.65	0.36%	0.38%
201	0.58	0.59	0.59	2.16%	2.20%
202	0.62	0.63	0.62	3.25%	3.28%
203	0.46	0.47	0.47	0.19%	0.19%
204	0.61	0.37	0.46	0.25%	0.15%
301	0.66	0.71	0.68	2.13%	2.29%
302	0.38	0.25	0.30	0.17%	0.11%
303	0.58	0.60	0.59	5.12%	5.31%
304	0.67	0.65	0.66	1.38%	1.34%
305	0.59	0.57	0.58	2.32%	2.22%
401	0.45	0.36	0.40	1.50%	1.21%
402	0.61	0.58	0.59	2.73%	2.60%
403	0.56	0.51	0.53	3.59%	3.25%
404	0.30	0.15	0.20	0.58%	0.28%
405	0.43	0.51	0.47	0.18%	0.21%
406	0.38	0.46	0.42	0.26%	0.31%
407	0.56	0.52	0.54	0.40%	0.38%
408	0.28	0.17	0.21	1.34%	0.79%
409	0.37	0.21	0.27	0.24%	0.14%
410	0.53	0.50	0.52	2.22%	2.08%
411	0.73	0.75	0.74	8.32%	8.53%
412	0.26	0.20	0.22	0.58%	0.45%
413	0.49	0.63	0.55	0.29%	0.37%
414	0.58	0.55	0.56	1.38%	1.32%
415	0.14	0.23	0.18	0.05%	0.07%
416	0.52	0.49	0.50	2.45%	2.35%
501	0.69	0.78	0.73	4.77%	5.35%
502	0.78	0.84	0.81	3.08%	3.32%
503	0.61	0.63	0.62	5.96%	6.11%
504	0.71	0.76	0.74	10.05%	10.76%
505	0.46	0.37	0.41	0.69%	0.55%
506	0.78	0.82	0.80	5.42%	5.72%
507	0.45	0.26	0.33	0.14%	0.08%
601	0.52	0.46	0.49	1.79%	1.57%
602	0.35	0.34	0.34	0.24%	0.24%
603	0.65	0.68	0.67	1.36%	1.42%
604	0.62	0.48	0.54	0.57%	0.44%
605	0.72	0.74	0.73	4.22%	4.33%
606	0.56	0.48	0.51	1.45%	1.23%
607	0.57	0.67	0.62	1.08%	1.25%
608	0.48	0.48	0.48	0.41%	0.41%
701	0.62	0.66	0.64	3.35%	3.59%
702	0.42	0.30	0.35	0.08%	0.06%
703	0.75	0.87	0.80	2.65%	3.07%
704	0.43	0.32	0.37	0.57%	0.43%
705	0.38	0.33	0.35	0.80%	0.69%
706	0.43	0.37	0.39	1.35%	1.16%

Table 8: Classification Results - Categories

Country	Accuracy	Precision	Recall	F1	N
Argentina	60.40%	0.48	0.48	0.46	3,586
Armenia	65.70%	0.65	0.55	0.65	207
Australia	73.01%	0.54	0.50	0.55	10,091
Austria	64.37%	0.48	0.50	0.54	2,366
Belgium	53.96%	0.43	0.42	0.42	26,091
Bosnia-Herzegovina	65.74%	0.45	0.46	0.48	3,193
Brazil	67.36%	0.56	0.54	0.55	2,255
Bulgaria	59.29%	0.42	0.47	0.46	926
Canada	60.74%	0.54	0.42	0.46	2,481
Chile	57.80%	0.39	0.48	0.45	4,810
Colombia	73.05%	0.53	0.56	0.56	590
Costa Rica	72.80%	0.57	0.58	0.57	10,570
Croatia	76.19%	0.62	0.60	0.65	2,579
Cyprus	58.37%	0.53	0.41	0.51	1,057
Czech Republic	62.46%	0.56	0.47	0.47	3,162
Denmark	61.80%	0.48	0.44	0.48	1,631
Ecuador	61.44%	0.58	0.55	0.60	638
Estonia	71.54%	0.66	0.61	0.68	889
Finland	66.06%	0.52	0.53	0.52	2,236
Georgia	62.41%	0.52	0.59	0.61	141
Germany	64.62%	0.55	0.51	0.52	10,650
Greece	53.55%	0.38	0.44	0.44	2,310
Hungary	69.87%	0.59	0.59	0.59	3,332
Iceland	67.58%	0.59	0.50	0.58	910
Israel	70.27%	0.57	0.57	0.57	1,238
Italy	72.38%	0.62	0.56	0.69	210
Latvia	76.67%	0.70	0.68	0.75	60
Lithuania	66.15%	0.50	0.44	0.46	7,595
Luxembourg	63.58%	0.54	0.52	0.52	2,938
Mexico	56.28%	0.41	0.43	0.40	4,636
Montenegro	64.86%	0.58	0.58	0.61	663
Netherlands	62.39%	0.48	0.44	0.46	6,932
New Zealand	62.11%	0.49	0.52	0.49	5,906
North Macedonia	75.61%	0.68	0.60	0.63	906
Norway	67.71%	0.50	0.44	0.47	8,383
Panama	71.29%	0.54	0.48	0.55	1,306
Peru	68.64%	0.55	0.60	0.56	5,090
Poland	66.61%	0.61	0.58	0.60	1,102
Portugal	65.56%	0.49	0.51	0.48	6,440
Romania	58.24%	0.51	0.48	0.51	1,147
Russia	60.49%	0.60	0.57	0.60	567
Serbia	65.32%	0.45	0.55	0.54	571
Slovakia	64.64%	0.56	0.52	0.55	1,120
Slovenia	52.88%	0.39	0.36	0.39	8,113
South Africa	71.21%	0.53	0.62	0.56	2,098
Spain	72.43%	0.60	0.60	0.61	11,986
Sweden	66.89%	0.53	0.48	0.51	4,111
Switzerland	60.82%	0.46	0.53	0.54	1,516
Turkey	63.91%	0.46	0.54	0.52	2,045
Ukraine	59.75%	0.48	0.48	0.50	395
United Kingdom	65.25%	0.52	0.51	0.51	2,501

Note: N relates to the number of quasi-sentences in the test dataset.

Table 9: Classification Results - Countries

Parfam	Accuracy	Precision	Recall	F1	N
10	58.91%	0.45	0.43	0.43	15,759
20	63.10%	0.52	0.51	0.50	19,603
30	65.09%	0.50	0.50	0.50	41,520
40	61.02%	0.49	0.48	0.48	31,121
50	62.63%	0.51	0.49	0.49	23,334
60	66.98%	0.52	0.54	0.52	29,326
70	65.45%	0.52	0.54	0.52	12,995
80	67.00%	0.59	0.57	0.62	303
90	71.15%	0.65	0.54	0.59	7,568
95	65.16%	0.51	0.46	0.48	3,304
98	73.40%	0.57	0.49	0.58	1,342
999	63.37%	0.63	0.72	0.71	101

Note: N relates to the number of quasi-sentences in the test dataset.

Table 10: Classification Results - Parfam

prob_estimates	accuracy	n(%)	cum_n(%)
> 95%	95.48%	15.30%	15.30%
90%-95%	86.40%	10.19%	25.49%
85%-90%	79.84%	8.06%	33.55%
80%-85%	73.76%	6.84%	40.39%
75%-80%	67.83%	6.36%	46.75%
70%-75%	63.79%	6.12%	52.87%
65%-70%	59.94%	6.02%	58.88%
60%-65%	55.33%	6.00%	64.89%
55%-60%	50.58%	6.25%	71.14%
50%-55%	47.59%	6.49%	77.63%
45%-50%	42.26%	6.27%	83.90%
40%-45%	38.23%	5.26%	89.16%
35%-40%	32.70%	4.30%	93.46%
30%-35%	28.79%	3.20%	96.67%
25%-30%	23.01%	2.07%	98.73%
20%-25%	21.22%	0.98%	99.72%
15%-20%	14.23%	0.26%	99.98%
10%-15%	16.28%	0.02%	100.00%

Table 11: Model Calibration - Probability Groups

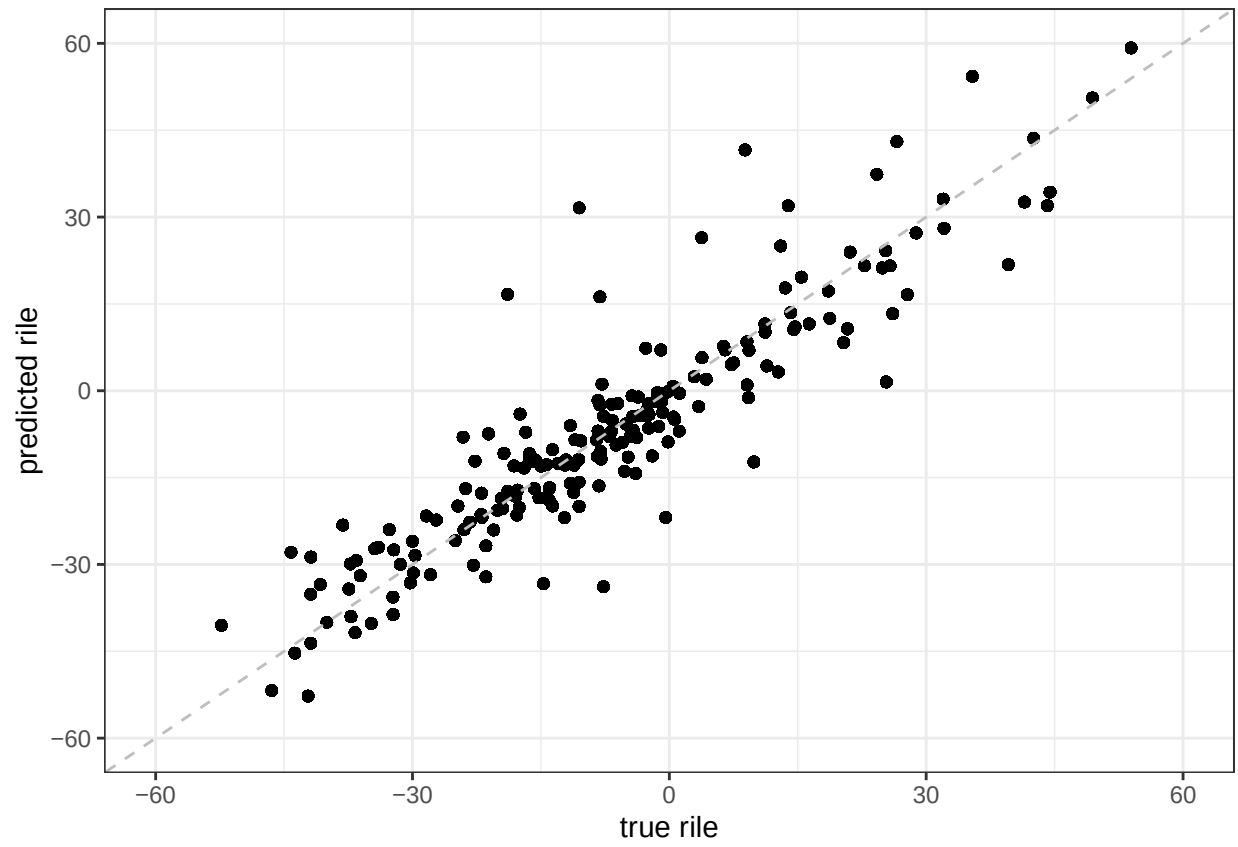


Figure 3: True rile values vs. predicted rile values

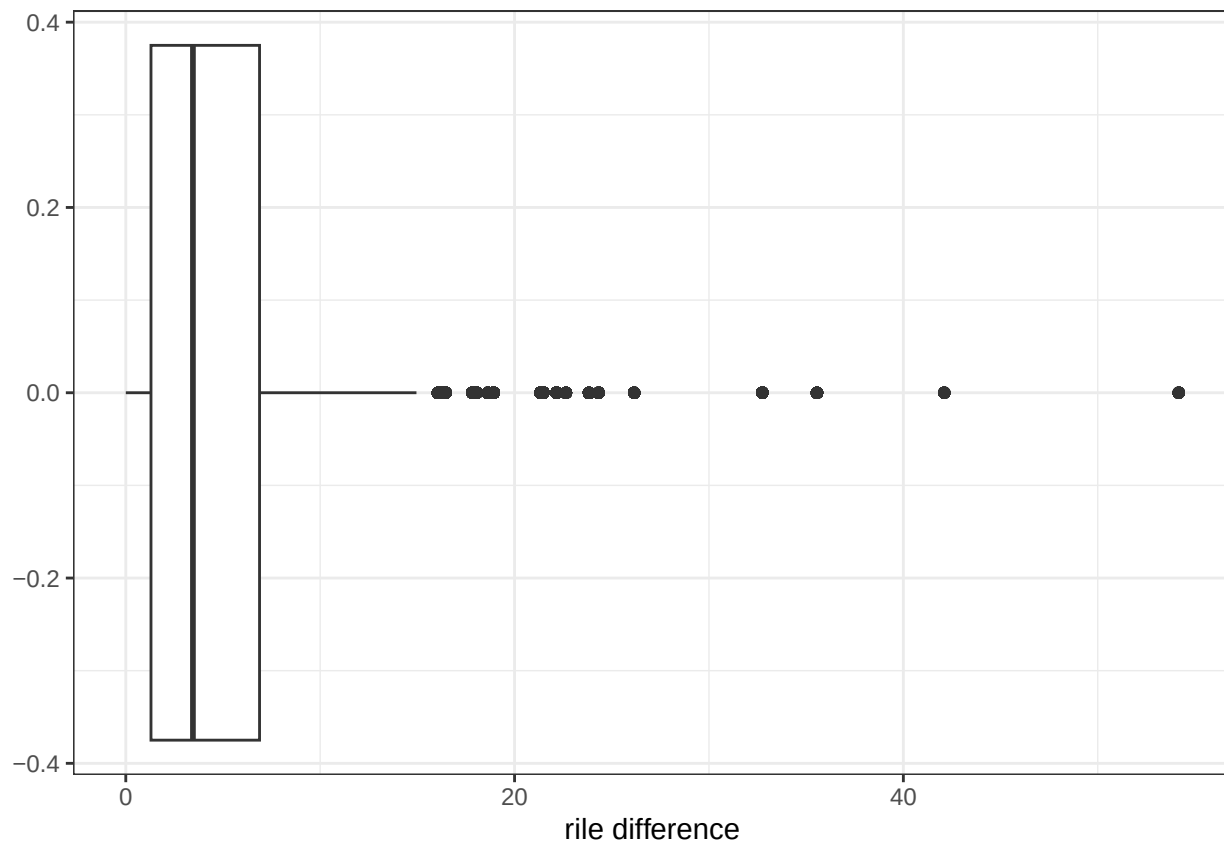


Figure 4: Absolute rile differences between true and predicted cmp codes